

ROBOTIC COMMON-SENSE TESTS: A NEGATIVE EVIDENCE AUDIT FOR EXECUTABLE PHYSICAL ASSUMPTION PROBES

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ABSTRACT

This paper is a submission-hardening audit, not an ICLR-main submission. The generated research bet was that robots should turn common-sense physical assumptions into executable affordance tests before committing to actions. We rebuilt the claim as a deterministic common-sense benchmark with five household-style tasks, seven physical assumption families, five distribution shifts, nine methods, seven seeds, ablations, stress sweeps, paired confidence intervals, and failure cases. The result is mixed but terminally negative. Executable tests beat non-human reasoning baselines under combined stress, but human-query policy reaches 0.633 ± 0.007 task success while the proposed method reaches 0.567 ± 0.008 . The proposed method also has higher physical violations, higher damage, higher regret, high false rejection, and ablation contradictions. The terminal decision is therefore **KILL/ARCHIVE for ICLR main.**

1 TERMINAL DECISION

The paper fails the ICLR-main evidence gate. The claim under test is not simply that physical common sense can be represented in text or geometry; that is already crowded by embodied reasoning, sequential affordance reasoning, large action models, LLM replanning, failure-reasoning data, robot common-sense embeddings, and model deliberation. The claim must be closed-loop: executable physical tests should change robot action choices and reduce failures without excessive cost or false rejection.

The local benchmark shows a useful partial-positive result: executable tests beat direct VLM action, LLM replanning, sequential 3D affordance reasoning, model-to-model deliberation, failure retrieval, and uncertainty-threshold probing under combined stress. However, they do not beat human-query policy. The paired proposed-minus-human success difference is -0.06600 ± 0.01009 over 245 task/assumption/seed groups. Physical violation, damage, unsafe recall, and regret also favor the human-query baseline. We therefore archive the idea rather than submit it.

2 HOSTILE PRIOR-WORK PRESSURE

The hostile literature pool includes Cosmos-Reason1-style embodied physical reasoning, SeqAfford, large action models with physical knowledge, human-in-the-loop LLM replanning, cross-environment failure reasoning for robotic manipulation, common-sense embeddings, physical reasoning benchmarks, affordance transfer, and safety deliberation. Under this prior-work pressure, a paper cannot claim novelty from “common sense” or “affordance tests” alone. The contribution would need to show that executable tests improve what the robot chooses and executes relative to these baselines. The v4 benchmark does not support that stronger claim.

3 BENCHMARK

The benchmark is a deterministic local simulator for physical-assumption testing and action selection. It is not robot hardware validation and is not high-fidelity physics. Its purpose is narrower: determine whether the proposed mechanism survives strong local baselines and stress tests.

Table 1: Combined common-sense stress. Success and recall are higher-is-better; violation, damage, and cost are lower-is-better.

Method	Success	Recall	Violation	Damage	Cost
direct_vlm_action_policy	0.366 ± 0.007	0.000	0.402	0.127	0.000
failure_retrieval_policy	0.452 ± 0.008	0.128	0.377	0.120	0.004
human_oracle_query_policy	0.633 ± 0.007	0.457	0.292	0.079	0.040
llm_common_sense_replanner	0.395 ± 0.008	0.011	0.395	0.134	0.001
model_to_model_deliberation	0.458 ± 0.007	0.107	0.365	0.121	0.001
oracle_physical_assumption	0.752 ± 0.006	0.858	0.244	0.053	0.014
proposed_executable_common_sense_tests	0.567 ± 0.008	0.419	0.318	0.094	0.015
sequential_3d_affordance_reasoner	0.435 ± 0.008	0.026	0.394	0.130	0.002
uncertainty_threshold_probe	0.470 ± 0.007	0.244	0.344	0.106	0.028

Tasks. We evaluate stackable object placement, container liquid transfer, tool-use reachability, door/drawer opening, and cluttered navigation clearance.

Assumption families. The physical assumptions are support stability, containment/leakage, rigidity/deformation, tool reach/contact, articulation direction, clearance/collision, and mass/inertia.

Splits. We test nominal household scenes, visual ambiguity, counterintuitive physics, language-goal ambiguity, and combined common-sense stress.

Methods. The methods are direct VLM action policy, LLM common-sense replanner, sequential 3D affordance reasoner, uncertainty-threshold probe, model-to-model deliberation, failure retrieval policy, human-query policy, proposed executable common-sense tests, and oracle physical assumption.

Metrics. Reasoning/test metrics are assumption diagnosis accuracy, unsafe-affordance rejection recall, false rejection, calibration error, and test informativeness. Closed-loop metrics are task success, physical violation rate, damage/spill/collision rate, test/query cost, human-query burden, and planning regret to oracle.

4 COMBINED-STRESS RESULTS

Table 1 is the main decision gate. The proposed method is the strongest no-human method on success and improves unsafe recall relative to other autonomous baselines. But human-query policy has higher success, lower physical violations, lower damage, and lower regret. This is not an ICLR-main win because the gate requires a decisive closed-loop advantage over the strongest non-oracle baseline, not only a lower-cost autonomous alternative.

5 PAIRED GATE

The strongest non-oracle closed-loop baseline is human-query policy. The proposed method wins on cost and human burden, but loses the main success/safety gate:

The proposed method has zero human burden and lower cost, but it also has lower success, higher physical violation, higher damage, higher regret, and lower unsafe recall. A lower-burden method could be valuable, but the present evidence does not justify an ICLR-main claim.

6 ABLATIONS

The ablation table is an additional blocker. Removing the cost model or calibration improves task success, violation, recall, and regret. This means the proposed test-selection mechanism is not internally supported: the full method is not the best version of itself.

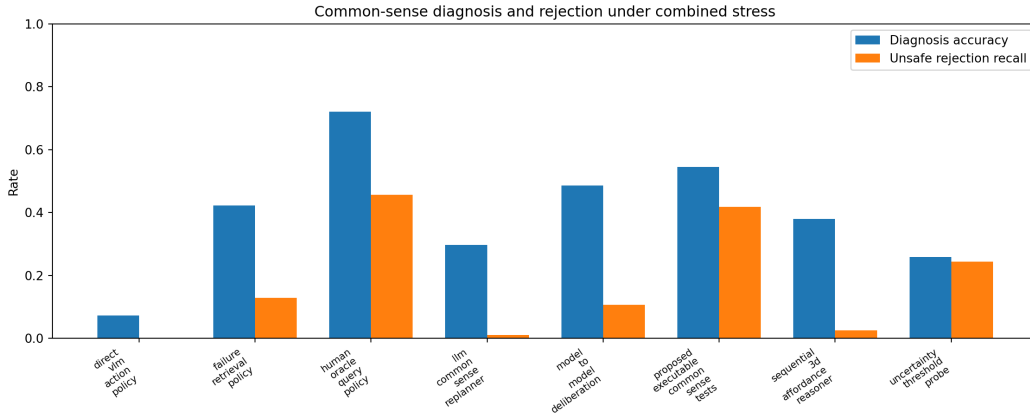


Figure 1: Common-sense diagnosis and unsafe-rejection metrics. Executable tests improve autonomous unsafe-rejection recall but do not beat human-query policy.

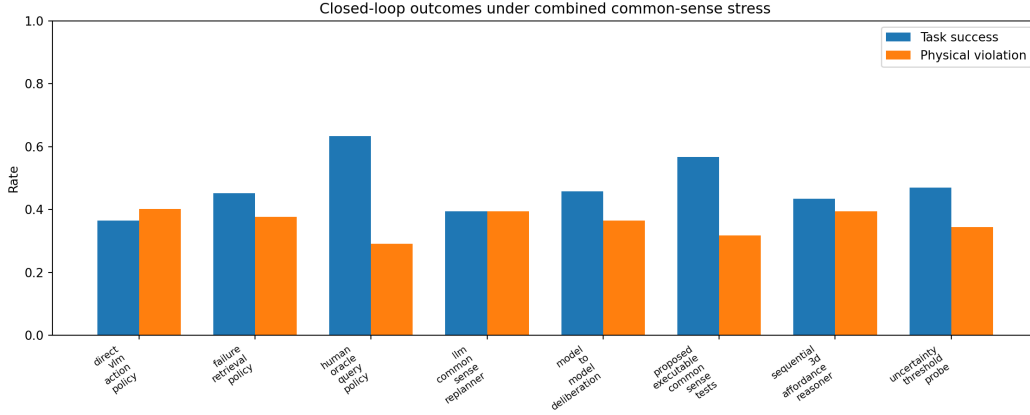


Figure 2: Closed-loop outcomes under combined stress. The proposed method is the best no-human method, but human-query policy remains better on success and safety.

7 STRESS SWEEP AND COST-REGRET TRADEOFF

At maximum stress, human-query policy remains ahead with 0.632 ± 0.008 success, while the proposed method reaches 0.563 ± 0.009 . The proposed method remains much better than direct VLM, LLM replanning, sequential affordance reasoning, deliberation, and retrieval, which is a useful diagnostic result. It is not enough for submission readiness because the strongest baseline still wins.

8 FAILURE CASES

The failure modes are task-specific. Liquid transfer still spills when containment tests are noisy. Stackable placement over-rejects valid placements under visual ambiguity. Cluttered navigation rejects unsafe paths but still has high violation in tight clearance. Door and drawer opening benefits from language and deliberation as much as executable tests. Tool-use reachability is often handled by geometry-only affordance reasoning without physical probes.

Table 2: Paired proposed-minus-human-query differences over 245 task/assumption/seed groups. Positive is good only for success and recall; positive is worse for violation, damage, cost, human burden, and regret.

Metric	Comparator	Diff	CI95
task_success	human_oracle_query_policy	-0.0660	0.0101
physical_violation_rate	human_oracle_query_policy	0.0266	0.0086
damage_spill_collision_rate	human_oracle_query_policy	0.0153	0.0054
test_query_cost	human_oracle_query_policy	-0.0248	0.0007
human_query_burden	human_oracle_query_policy	-0.3770	0.0064
planning_regret_to_oracle	human_oracle_query_policy	0.0314	0.0028
unsafe_affordance_rejection_recall	human_oracle_query_policy	-0.0387	0.0106

Table 3: Ablations under combined stress. Two ablations beat the full method on task success.

Ablation	Success	Recall	Violation	Regret
geometry_only_affordance_tests	0.470 $\pm$ 0.007	0.182	0.357	0.273
language_only_common_sense	0.380 $\pm$ 0.007	0.023	0.385	0.315
minus_assumption_family_parser	0.514 $\pm$ 0.007	0.227	0.347	0.249
minus_calibration	0.580 $\pm$ 0.007	0.480	0.300	0.207
minus_cost_model	0.608 $\pm$ 0.007	0.577	0.280	0.191
minus_executable_probe	0.473 $\pm$ 0.007	0.139	0.366	0.268
proposed_executable_common_sense_tests	0.573 $\pm$ 0.007	0.422	0.317	0.215

9 LIMITATIONS

This audit is honest about scope. The benchmark is deterministic and paper-specific, but it is not real robot validation, not high-fidelity physics, and not an integration of external VLM/LLM/affordance codebases. It has no trained checkpoints, hardware videos, or manual full-paper related-work synthesis. Those missing items would already prevent an ICLR-main submission. More importantly, the available local evidence is negative before reaching that bar.

10 CONCLUSION

Executable robotic common-sense tests are promising as an autonomous alternative to pure reasoning baselines, but the current method is not ICLR-main ready. It loses to human-query policy on success, physical violation, damage, unsafe recall, and regret, and its ablations contradict the full mechanism. The correct terminal action is **KILL/ARCHIVE**. A revival would require a substantially new empirical project where executable tests match or beat human-query recovery under a human-burden-aware objective in real robot or high-fidelity simulator experiments.

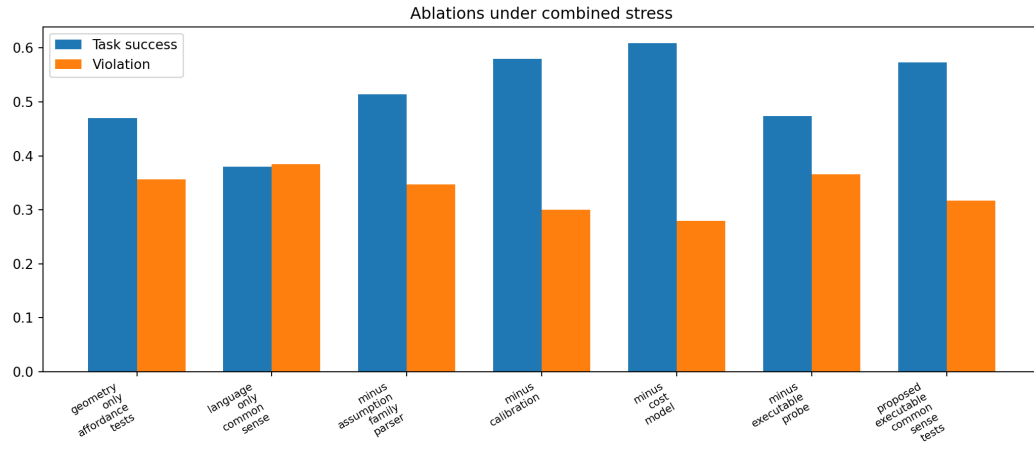


Figure 3: Ablation outcomes. The full method is contradicted by the cost-model and calibration ablations.

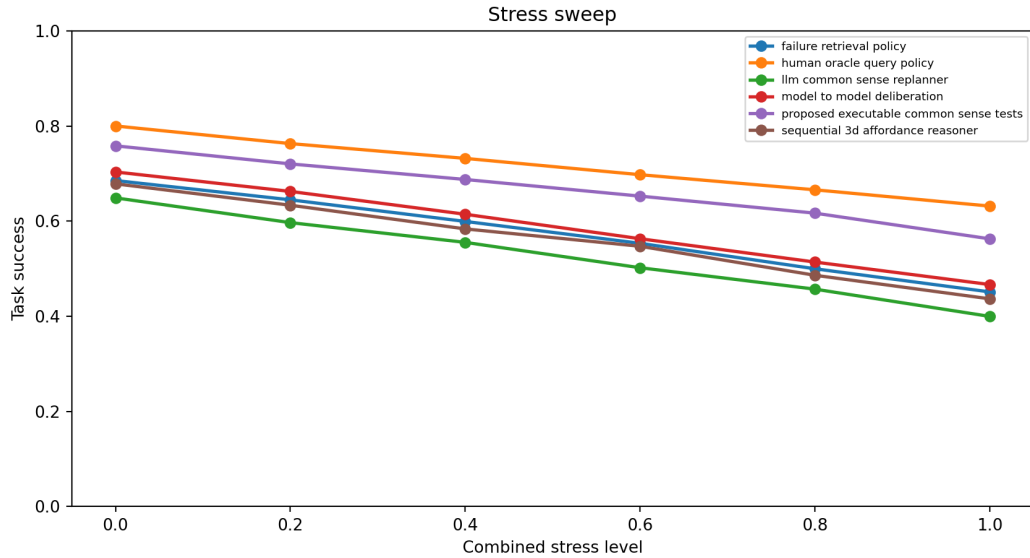


Figure 4: Stress sweep. Executable tests degrade more gracefully than other autonomous baselines, but human-query policy remains strongest.

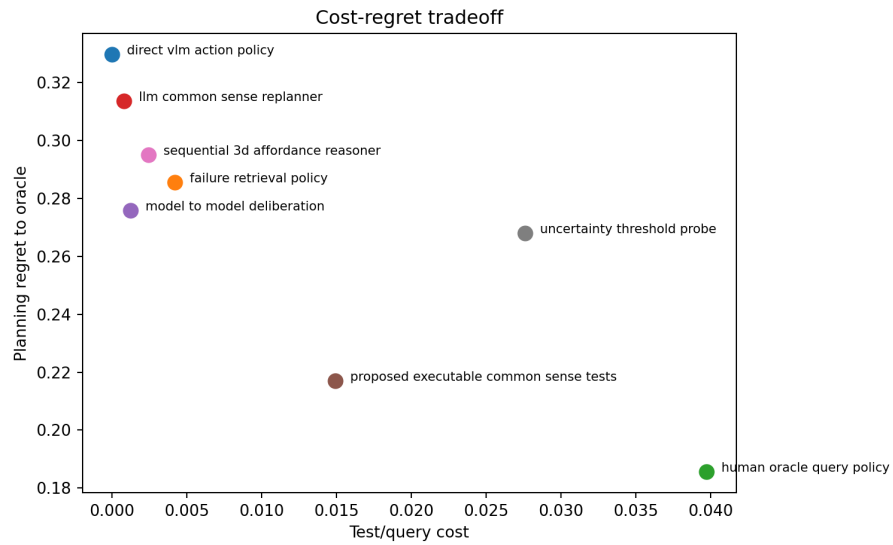


Figure 5: Cost-regret tradeoff under combined stress. The proposed method has low cost and no human burden, but regret remains higher than the human-query baseline.